PCDSMetadataChanges

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20250515 - Ted Weick, James Hiebert, Francis Zwiers

- Concatenation could be broken down in to 3 types
  - Easy - record is continuous across 2 sites (one site ends at hour x, other site starts at hour X+1)
  - Moderate - there is some overlap, but the overlap agrees between both sites
  - Hard - there is some overlap, but the overlapping data is different.
- AGRIWeather - data. At some point ECCC will be polling and posting AGRIWeather data on Datamart. Plan is to wait until that data is posted, then request a backfill from Stephanie tam on all historical data.
- FERN data will be available for ingest at the year end. Ted arrange to send complete record of data to upload (likely via a BC Box location that only James and I have access to).
- Tong Li will be asked to start the work on each of the networks - starting with Data Loading.

Actions

Ted

☑ Clearly note which records need to be deleted (delete and Short Records) vs other actions	
☑ to briefly review the MOTIm network to see if there are problems - As far as I can see there are no problems with the metadata.	
☑ Provide James with a list of sites that may be relevant to PCIC for Alberta, NWT, and Yk	Check out

c2b00882	Maligne	AB	
c2b01bf4	Lake Louise	AB	
cba4d0fc	Bryant	AB	
cba54764	Windy	AB	
cba51718	Virginia Falls	NW	
cba24550	Saskatchewan Crossing	AB	In spreadsheet
cba06648	Waterton Townsite	AB	
cba096cc	Summitt Lake		
c2b14972	Castle Mountain	AB	
c2b058fe	Ranger Creek	AB	
cba164b2	Dorothy	AB	

☑ Facilitate contact information between James and Carmen Riddle for Snow Attribution list. -	Email sent
☑ Change in station operations - Brenda Mines and Moyie Mountain	- Created new records for each site transferred and requested moving of data collected from October 1st, 2022 onward to the new site/resp combination.
☑ Sooke Dam -	Ted to check for EMS Ids and where those locations are. Data has been tagged to keep with modifications or to be deleted.

The various Sooke Fireweather stations have different EMS IDs - from EMS, those sites are the following:

EMS ID	NAME	OFFICE	OFFICE NAME	ESTABLISHING AGENCY	EST. AGENCY NAME	PURPOSE DESCRIPTION	LATITUDE (Decimal Degrees)	LONGITUDE (Decimal Degrees)	ELEVATION	ESTABLISHED DATE	CLOSED DATE	COMMENT	NUMBER OF SAMPLES	FIRST COLLECT. START DT	LATEST COLLECT. START DT
E206589	SMITHERS ST JOSEPHS SCHOOL	60	SKEENA, SMITHERS	60	Skeena	TREND	54.783308	127.177324	495	18-Feb-86			1233	28-Jan-86	06-Feb-08
E316070	CROFTON ELEMENTARY	90	VICTORIA	90	Headquarters, Victoria	BACKGROUND	48.86	123.643889	38	01-Apr-19			0		
E309527	FORT ST. JOHN NORTH CAMP C	BCH	BC Hydro	90	Headquarters, Victoria	BACKGROUND	56.203289	120.90108	584	23-Aug-17			0		
E312331	PRINCE RUPERT FAIRVIEW	60	SKEENA, SMITHERS	90	Headquarters, Victoria	BACKGROUND	54.292615	130.352188	15	04-May-18			0		

☑ Each site identified as a potential site to investigate data overlaps.	

☑ Follow up on MVRD sending data to DataMart and Correct information for Capilano Fire weather site.	☑ Asked Kristine C about status - indicates its on the list to do, but not yet ☑ Sent an email to Pete Marshall at MVRD - wondering about time line to post water services data to datamart.
☐ Add NAV Canada sites to the PCIC list.	This might have to be a separate task
☐ FINALIZE ORG NAMES	There are two spreadsheets - one with the new org names, and one with the old. Preference is to use the new org names
☑ Provide James with the CRD data for the noted missing information.	*Loaded to Share point site*
☑ Put the spreadsheet in a group accessible site and grant James access once he has BC EID	
☑ McBride Peak	-> Sunbeam
☑ To remove Rio Tinto duplication -	**stations and data reported through BC ENV to be deleted from the PCDS. Note there is a request to Rio Tinto for data to fill data gaps on the PCDS**

James

☑ Provide a list of sites within FERN North that he has new data for.	.r.t. what FERN data we have yet to be inserted, we have data from Vanessa for 2023 and 2024. The sites (according to the filenames) are as follows: Atlin school, Barren, Blackhawk, BoulderCr, Bowron Pit, Bulkley PGTIS 1, Canoe, Caribou Pass, Chapman, ChiefLk, Coalmine, CPF PGTIS 3, Crooked Lake, CrystalLk, Dennis, Dunster, Endako, George, Gunnel, Hourglass, HudsonBayMtn2, IBB2 Ganokwa Canyon, IBB3 Pine Creek, Inklin, Kluskus, MacJxn, Middlefork, NondaWx, North Fraser, PinkMtnWx, SaxtonLakeWx, Seebach, Stone Creek, Sunbeam, Tamarac, Thompson, Willow, Bowron PGTIS 2 She also provided the station location spreadsheet that, at first glance, appears to contain everything for all of those stations.
☑ Provide Ideal Format list.	As for the ideal format for data infilling, we can work with anything as long as it's consistent. The ideal, is one record per weather observation with the following attributes: [time value variable_name unit network_name native_id lat lon elevation] If some things are missing (unit, lat lon, elevation), we can just match it up with metadata that we already have. But again, that's the *ideal*... we can and have worked with whatever you can get us, given time and availability of resources. From <https://outlook.office.com/mail/AAMkAGVhODJmODBiLTcwYjYtNGFkMC05ZWUzLTVjNTE3MTJmNjhhNgAuAAAAAADPPjaAKAfTEZvvAIBfFSnoAQDNqqDFb%2F8oTaatwXE%2FRQHwAADevyXOAAA%3D/id/AAQkAGVhODJmODBiLTcwYjYtNGFkMC05ZWUzLTVjNTE3MTJmNjhhNgAQABLtGvVQFvhIlgKDNebAvao%3D?nativeVersion=1.2025.829.200>

Network	Recommendation	
Priorities	1. Attribution - Network name adjustments 2. Attribution - Deletion if appropriate 3. Additional - New stations that are not on the PCDS list **or** modify existing records where tombstone information is not present 4. Tombstone data - adjustments to name, ID, location as identified. Modify BOT to read in elevation from SWOBML and report out if there is a significant change in lat/long/elevation for further review by Climate Network Coordinator. 5. Concatenation - Amalgamating data streams from orphaned data lines. 6. Identify and report out on data gaps - seek missing data from source agency. 7. Delete stations with short records. Modify append code to incorporate elevation from SWOBML. Ted is still working on ORG name changes.	Questions 1. What should be the practice when a station is transferred to another agency? 2. What should be the practice when two agencies collaborate on one site and report data for that site (data sets may be different) 3.

Network names			New Name					
Old Designation					PCIC PCDS Naming			
Agency	Network	Transitionary Designation	Agency	Network Name	Current	suggested Network name (blanks infer no change)	DataMart Partner Directory *(italics and Rjustified planned)*	
AF	AGRI	BC AgriWeather	BC Gov	BC AGRIWeather	AGRI	BC AGRIWeather	*bc-agriweather*	
ENV	AIR	ENV/AIR		Air	ENV-AQN	BC-Air	bc-env-aq	
	ASWS	ENV/ASWS		Snow	ENV-ASP	BC-Snow	bc-enc-snow	
	Hydrometric	ENV/WSC		Hydrometric				
	Climate	ENV/CLIM		Climate				
FOR	BCTS	FOR/BCTS		Timber Sales	FLNRO-WMB	BC-TS	bc-forestry	
	BCWS	FOR/FWx		Fire Weather		BC-FWx		
	FERN	FOR/FERN		Forest Ecosystem Research	FLNRO-FERN	BC-FERN		
MOTI or TRAN	Ministry of Transportation and Infrastructure	MoTT		Avalanche & Road Weather	MoTIc / MoTIm	BC-TRAN	bc-tran	Not Transferred
ARDA	Agriculture Research Development Network			Agriculture Research Development Network	ARDA			
FRBC	Forest Renewal BC			Forest	FRBC			
BC Hydro	Hydrometeorologic Monitoring and Data Management	BCH/GSO-HMP	BC Hydro	GSO-HMP	BCH		*bc-hydro*	
	Site C	BCH/SITEC						
BC Hydro/Parks	Hydrometeorologic Monitoring and Data	Parks Canada	Canada	Parks				
				Parks Fire		PC-FWx	pc-firewx	
ECCC	Daily Climate	ECCC/DAILYCLI		MSC Daily	EC / EC RAW			
	HYDAT/CLIM	ECCC/HYDATC		No longer required				
	MARINE/Moored	ECCC/BUOY		MSC Buoy				
	SURFACE/CLIMATE	ECCC/SFCWx		MSC Surface				
CRD		CRD-WS	CRD	Water Supply	CRD	CRD-WS	bc-crd	
FOR-UNBC	University of Northern British Columbia/Cariboo Alpine Mesonet	UNBC/CAM	UNBC	Cariboo Alpine Monitoring	UNBC_CAM			
MVRD	Air Quality	MVRD/AIR	MVRD	Air	Mvan	MVRD	bc-mvrd	
	Watershed Management Network	MVRD/WT		Water Supply				
NAVCAN	NAVCanada	Nav Canada	NAVCAN	NAVCAN				
RT	Rio Tinto	RT	Rio Tinto	Water Supply	RT		bc-RioTinto	
HAKAI		HAKAI	HAKAI	HAKAI	HAKAI			
			Prince Rupert Port Authority	PRPA		PRPA		

CRD	○ Discontinue the new sites, and move the data to the older sites - there should only be one metadata record per reporting station - Assumption is that PCIC will have to amend ingest code ○ Input complete dataset for Judge and Rithet Creeks ○ PCIC to initiate data downloads from Datamart for Judge Creek HY002	Data to fill gap will be provided under separate note.
BC AGRIWeather	• Attribution: Delete any MVRD data held under the AGRI org in PCDS • Additional: Create the 68 new sites and either start polling data or wait until data is posted on Datamart and refill PCS from a query from the BC AGRIWeather database to the date of code initiation. • Concatenation: Amalgamate the data from the 22 identified stations into one station record (each station) under the BC AGRIWeather banner	Note - there are 18 stations where the data should remain in the PCDS but not publicly available.
BC Transportation	• Additional: Add new stations to PCDS, modify append code to incorporate the new stations from Datamart. Get the historical record (such as it exists) for data ranging for site setup to the time of the BOT adjustments. • Delete SOOKE DAM FWx - This is a common metadata site name that appears across FWx, MoTT, Snow programs.	
Rio Tinto	• Update Location information • Missing data from December 1st, 2019 through to mid December 2024. For 5 of the stations. Eutsuk has a larger gap going to 2013.	Data has been queried from Rio Tinto.
BC FWx Network	• Attribution: Delete the indicated misattributed data **except the Parks Canada sites** **Note MVRD stations @ Seymour, Capilano and Coquitlam Fire weather sites are no longer reported through FWx Datamart (effective 20250912).** • Additional: Add the Indicated Sites. • Update names at indicated locations • Remove Suffix and prefix "ZZ" from site names. This is only an internal marker for Wildfire staff. These changes have not been marked in the spreadsheet. You may want to delay this change until the short record stations have been deleted. • Update Locations/Elevations for indicated stations • Delete the Site data and metadata for any sites with short records (duration less than a year) except for those currently reporting.	SPECIAL NOTE: PCIC Station ID 2240 and 13000 (Cariboo Ck and Cariboo Creek respectively) are essentially the same station, however they separately defined within the FWx Database. Therefore they should be maintained as separate metadata sites within the PCDS.
BC FERN	Queries for updated data sets have been sent to data providers (except Bill Floyd). Expected return date is November, December of 2025 and January of 2026. Ted - Communicated desired format to Brian, Sheena, Natasha McBride Mountain (12063) has been renamed Sunbeam (13069). Check for data gaps under Sunbeam and see if they can be filled under McBride Mountain. Once gaps have been filled delete McBride Mountain (12063).	
Snow Program (BC Hydro and ENV - Not MVRD, Other agencies).	• Attribution: Data collator (ENV) has a requirement to provide data from multiple networks to core data users. Hence Attribution to ENV is occurring for stations that are not operated by ENV. ENV now has an attribution file. Some data should be deleted but caution is required before doing this - check that the data coverage is as continuous as possible in the source agency site, before deleting the misattributed site. • One site (Johanson Lake, 13217 and13128) seems to have two metadata records. Data needs to be amalgamated.	Attribution needs to be checked against attribution file.

Air Program (MVRD and ENV, PRPA, BC Hydro Site C).	• Make corrections to the Tombstone information where indicated (Attribution, ID, Name, Location)
	• Rejoin the 5month window missing data for each of the station.
	• Any and all data collected through ENV Air should attributed to the appropriate agency. In this case do not delete the data unless there is already data in place from the source agency
	  ○ Test for the presence of data under two networks from that specific site.
	  ○ If duplicate data exists, keep the data that came from the operating agency.
	• Upload and replace any existing Meteorological data from ENV Air Program that is posted on the finalized	

	network, once the okay is given from the Air Data Administrator. Create a process to ensure that this update occurs once annually from the previous year's data (usually available after June of the following year).	
Parks Canada sites	• All data available from https://dd.weather.gc.ca/today/observations/swob-ml/partners/pc-firewx/ BOT Needs to look at this information BC and surrounding points. • Existing data (from FWx) needs to be reattributed to Parks Canada, location information needs updating.	